[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

SAW, BAND, CUTOFF, HORIZONTAL, AND COMBINATION HORIZONTAL AND VERTICAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers manual and semiautomatic horizontal bandsaws with mitering capabilities used for metal cutoff operations. This CID includes both hinge-type multistep and infinitely variable-speed horizontal bandsaws and twin post-type infinitely variable-speed horizontal bandsaws. This CID also incorporates the combination horizontal and vertical multistep speed bandsaws. The bandsaws covered by this CID are intended for general cutoff operations on metal stock.

2. CLASSIFICATION. The bandsaws shall be of the following types, styles, and sizes. The type, style, and size to be furnished shall be as specified (see 7.2(b)). The following sizes refer to the maximum bar stock capacity the machine can accommodate:

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AMSC N/A FSC 3405

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Type I - Horizontal

Style A - Hinged

Size 1 - 5 by 10 inches
Size 2 - 6 by 10 inches
Size 3 - 6 by 12 inches
Size 4 - 8 by 10 inches
Size 5 - 8 by 12 inches
Size 6 - 8 by 16 inches
Size 7 - 8 by 24 inches
Size 8 - 10 by 16 inches
Size 9 - 12 by 12 inches
Size 10 - 12 by 16 inches
Size 11 - 12 by 20 inches

Style B - Twin post

Size 1 - 9 by 17 inches
Size 2 - 12 by 12 inches
Size 3 - 12 by 16 inches
Size 4 - 12 by 20 inches
Size 5 - 14 by 17 inches
Size 6 - 15 by 20 inches
Size 7 - 24 by 24 inches
Size 8 - 32 by 32 inches
Size 9 - 40 by 40 inches

Type II - Combination horizontal and vertical

Size 1 - 3 by 6 inches
Size 2 - 6 by 10 inches

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing.

3.2 Components. The bandsaw shall include, as a minimum, the following components:

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3.2.1 Base. The base shall support all components of the horizontal bandsaw and shall be constructed with the strength and rigidity necessary to meet the performance requirements shown in table I. When specified (see 7.2(c)), the base of type I size 1 through 3 machines shall be equipped with wheels for portability. All other sizes covered herein shall have bases that provide means for leveling and securely anchoring the machine.

3.2.2 Saw frame. Unless otherwise specified (see 7.2(d)), the saw frame shall provide manual blade tensioning and shall be equipped with a blade tension indicator. Type II machine saw frames shall be capable of being locked in a vertical position.

3.2.3 Blade drive system. Unless otherwise specified (see 7.2(e)), the bandsaw blade speed and cutting force shall be that considered as standard by the manufacturer. The bandsaw shall be equipped with indicators for displaying blade speed and cutting force. The blade shall stop at the completion of a cut and shall provide an automatic stop function in the event of blade breakage.

3.2.4 Automatic material-handling system. When specified (see 7.2(f), an automatic material-handling system shall be provided. The automatic material-handling system shall include an automatic vise, adjustable material stop, nesting fixtures, and loading and discharge worktables as specified (see 7.2(g)).

3.2.5 Blade cleaner. Blade brushes shall be provided in the saw frame to remove metal chips from the saw blade.

3.2.6 Bandsaw blades. When specified (see 7.2(h)), saw blades of the specified material, quantity, size, tooth pitch, and type shall be furnished with the bandsaw.

3.2.7 Controls. Machine controls shall be convenient to the operator. Unless otherwise specified (see 7.2(i)), the manufacturer's standard control system shall consist of push buttons and selector switches for start/stop, emergency stop, blade speed, cutting force, blade tension, and vise clamping force. When provided, bandsaws with semiautomatic capabilities shall allow the operator to preset the blade speed and cutting force and then start the sawing process. At the end of the cut the process shall cease automatically.

3.2.8 Coolant system. When provided, the bandsaw blade coolant system shall consist of a sump, power-driven pump, and coolant nozzles(s). The coolant nozzle(s) shall be adjustable to allow coolant to be directed to the cutting area. Means shall be provided to determine and maintain coolant level within the sump. A removable strainer shall be provided to remove metal chips from the coolant.

3.2.9 Hydraulic system. When furnished, the hydraulic system shall provide a fluid level indicator and over-pressure protection.

3.2.10 Electrical system. Unless otherwise specified (see 7.2(j)), the electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79. The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability

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and shall be wired for the voltage specified (see 7.2(k)). Unless otherwise specified (see 7.2(l)), the control system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply. The electrical circuitry shall provide low voltage and overload protection.

3.2.11 Electric motors. The horizontal bandsaw motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see 7.2(m)), the motor(s) shall have open dripproof enclosures. The bandsaw drive motor shall have a 30-minute duty cycle horsepower rating,

3.2.12 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(n)).

3.3 Performance. Unless otherwise specified (see 7.2(o)), the horizontal bandsaws shall be capable of producing cuts in SAE 1045 steel that comply with the requirements shown in table I when tested in accordance with sections 5.3.1 through 5.3.4.

TABLE I. Horizontal bandsaw performance requirements.

Machine classification	Cutting rate, square inches per minute	Perpendicularity, inch per inch of workpiece diameter/height
Type I, style A, sizes 1 – 8 Type II, sizes 1 and 2	4	0.004
Type I, style A, sizes 9 – 13 Type I, style B, sizes 1 – 9	15	0.002

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(p)).

3.5 Safety and health requirements. The horizontal bandsaw shall be designed and manufactured in accordance with American National Standards Institute (ANSI) B11.10. The manufacturer shall ensure the machine and all equipment and accessories used on the machine shall comply with Occupational Safety and Health Association (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI B11.10 and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(q)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(r)), measuring and indicating devices shall be graduated in both the US and SI systems of measurement.

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3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(s)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(t)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user’s health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The horizontal bandsaw shall be inspected for compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(u)), the horizontal bandsaw shall be tested in accordance with sections 5.3.1 through 5.3.4.

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5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(v)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(w)). Unless otherwise specified (see 7.2(x)), the machine shall be tested under the normal environmental conditions of the manufacturer's facility.

5.3.2 Operational test. The horizontal bandsaw shall be operated in accordance with the manufacturer's standard operating procedures. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests for type I, style A, sizes 1–8; and type II machines. The bandsaw shall be tested for cutting rate compliance. The test workpiece shall be an SAE 1008 steel plate with dimensions of 1/2 inch thick by 6 inches wide by 12 inches long. Three test cuts shall be made. When in the vertical configuration, the type II bandsaw shall cut along a 4 1/2-inch scribed radius semicircle in an SAE 1010 steel plate that is 1/4 inch thick by 6 inches wide by 6 inches long without deviating more than 1/8 inch at any point. The choice of saw blade, coolant, blade speed, and blade force for all bandsaw tests shall be the option of the supplier.

5.3.4 Performance tests for type I, style A, sizes 9–11; and style B bandsaws. The bandsaw shall be tested for cutting rate compliance. Either a round or rectangular workpiece may be used for the test. The test workpiece shall have a 5-inch diameter or a 5- by 5-inch cross-section. Three test cuts shall be made. The choice of saw blade, coolant, blade speed, and blade force for all bandsaw tests shall be the option of the supplier. The test results will be compared to the performance requirements shown in table I.

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(y)).

5.4 Acceptance. Unless otherwise specified (see 7.2(z)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. The results of the performance tests shall be compared to the performance requirements shown in table I. Failure of the machine to meet the performance requirements shown in table I for the performance category selected shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.2(aa)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of referenced documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B11.10 Safety Requirements for the Construction, Care, and Use of Metal Sawing Machines

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

National Fire Protection Association (NFPA)

ANSI/NFPA-79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

National Electrical Manufacturers Association (NEMA)

NEMA MG-1 Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturer's Association, 1300 North 17th Street, Suite 1847, Roslyn, VA 22209.

Occupational Safety and Health Administration (OSHA)

OSHA 1910 Occupational Safety and Health Standards for General Industry (29 CFR 1910)

Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, style, and size of bandsaw required (see 2).
c. Wheels, if required (see 3.2.1).
d. Saw frame, if different (see 3.2.2).
e. Blade speed and cutting force, if different (see 3.2.3).
f. Automatic material-handling system, if required (see 3.2.4).
g. Automatic vise, adjustable material stop, nesting fixtures, and loading and discharge worktables, as specified (see 3.2.4).
h. Saw blade material, quantity, size, tooth pitch, and type, if required (see 3.2.6).
i. Control system, if different (see 3.2.7).

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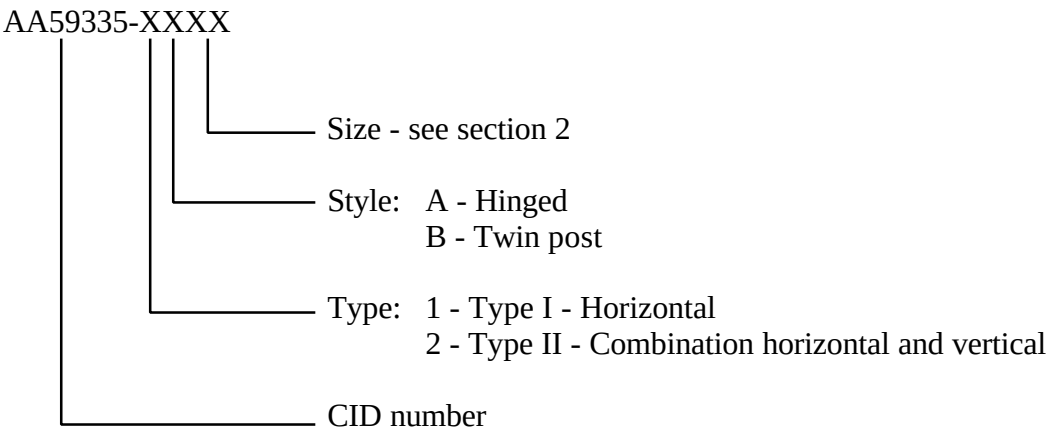
j. Electrical system industrial standard, if different (see 3.2.10).
k. Electrical system voltage, as specified (see 3.2.10).
l. Control system voltage, if different (see 3.2.10).
m. Motor enclosure, if different (see 3.2.11).
n. Optional accessories, as specified (see 3.2.12).
o. Performance, if different (see 3.3).
p. Machine dimensions and weight, as specified (see 3.4).
q. System of units required (see 3.6).
r. Dual system of units, if required (see 3.6).
s. Nameplate, if different (see 3.7).
t. Lubrication plate or chart, if required (see 3.8).
u. Testing, if different (see 5.3).
v. Test responsibility, if different (see 5.3.1).
w. Certification of calibration, as specified (see 5.3.1).
x. Test environmental conditions, if different (see 5.3.1).
y. Optional tests, as specified (see 5.3.5).
z. Acceptance, if different (see 5.4).
aa. Packaging requirements, as specified (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

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